

LEO ZHOU

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Education

Rice University | Bachelor of Science in Computer Science | **GPA: 3.92/4.00** **Expected Graduation, May 2028**

• **Relevant Coursework:** Data Structures & Algorithms, Computer Organization, Discrete Math, Linear Algebra

Technical Skills

Languages: Python, C/C++, Go, JavaScript/TypeScript, Swift, SQL, HTML/CSS

Frameworks & Libraries: FastAPI, Flask, React, Node.js, SwiftUI, PyTorch, ONNX, REST APIs, WebSockets

Tools & Cloud: AWS (ECS, SQS, RDS), Docker, Terraform, PostgreSQL, Redis, SQLite, Git, GitHub Actions

Experience

Software Engineer Intern | **CanSemi**

June 2026 – Present

- Engineered an event-driven parser in Python extracting test measurements from 500+ binary test equipment (ATE) logs
- Designed a normalized PostgreSQL schema and batch ETL pipeline loading 40 wafer lots (~12,000 records) in <2 min
- Developed a REST API in Python and FastAPI with 8 endpoints, cutting report turnaround ~75% for 6 engineers
- Built an interactive data dashboard in React with hover and drill-down, replacing a manual Excel reporting process

Software Engineer | **Inkstone Technologies**

May 2026 – Present

- Shipped a native iOS reading application in Swift and SwiftUI that adapts content to each user's known vocabulary
- Integrated Apple Vision OCR into an on-device scanning pipeline, overlaying pronunciation on unfamiliar words only
- Wrote a Python pipeline to parse, normalize, and frequency-rank a 113,000-entry dictionary into an indexed SQLite DB
- Designed a local-first SQLite layer on the raw C API, driving spaced-repetition review and a WidgetKit widget

Embedded Software Engineer | **Rice Wind Energy**

October 2025 – April 2026

- Programmed ESP32 microcontroller firmware in C/C++ to monitor turbine speed, power output, and blade pitch
- Implemented a six-state safety state machine with fault detection and recovery, plus a peak power tracking algorithm

Projects

BlunderNet — Self-Training Chess Engine  | *Python, PyTorch, C++, pybind11, GitHub Actions*

- Developed an AlphaZero-style neural network (residual CNN with policy and value heads) in Python and PyTorch
- Accelerated Monte Carlo Tree Search by 1.8× with a C++ implementation bridged to Python via pybind11
- Automated the self-play training loop with GitHub Actions CI/CD, from data generation to model evaluation
- Benchmarked model strength with held-out move prediction, a 1,200-puzzle test suite, and matches vs. Stockfish

BlunderNet Arena — Cloud Serving Platform  | *Go, AWS, Terraform, Docker, Redis, React*

- Built a site where anyone can play the BlunderNet engine, on a stateless Go API fleet behind an AWS load balancer
- Moved engine inference to autoscaling ONNX workers behind AWS SQS, keeping play responsive when traffic spikes
- Streamed live game moves to browsers over WebSockets, fanned out with Redis pub/sub across every API instance
- Provisioned the AWS environment (VPC, ECS, RDS, SQS) in Terraform, rebuilding the whole stack from one command
- Load-tested with k6 over live WebSocket sessions at zero failed requests, with API responses under 30 ms at p99

SeatLive — Concurrency-Safe Booking System  | *Python, FastAPI, PostgreSQL, Redis, WebSockets, Docker*

- Built a ticket booking backend in Python and FastAPI that prevents double-booked seats using atomic transactions
- Implemented idempotent request handling with client-supplied keys so network retries never create duplicate bookings
- Validated correctness with CI tests simulating 50 concurrent users and Locust load tests at 4,459 requests, zero failures

Leadership & Athletics

- **New Zealand Junior National Fencing Team** — Captain, Coach, Referee; 30+ titles, Junior World Championships
- **Rice Fencing Club** — Manager · **Physics Pathfinders** — President; student-led charity, 600+ students, 22 schools